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StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-.

Physiology, Thyroid Hormone

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Last Update: June 5, 2023.

Introduction

The thyroid hormone is well known for controlling metabolism, growth, and many other bodily functions. The thyroid gland, anterior pituitary gland, and hypothalamus comprise a self-regulatory circuit called the hypothalamic-pituitary-thyroid axis. The main hormones produced by the thyroid gland are thyroxine or tetraiodothyronine (T4) and triiodothyronine (T3). Thyrotropin-releasing hormone (TRH) from the hypothalamus, thyroid-stimulating hormone (TSH) from the anterior pituitary gland, and T4 work in synchronous harmony to maintain proper feedback mechanisms and homeostasis. Hypothyroidism, caused by an underactive thyroid gland, typically manifests as bradycardia, cold intolerance, constipation, fatigue, and weight gain. In contrast, hyperthyroidism caused by increased thyroid gland function manifests as weight loss, heat intolerance, diarrhea, fine tremor, and muscle weakness.

Iodine is an essential trace element absorbed in the small intestine. It is an integral part of T3 and T4. Sources of iodine include iodized table salt, seafood, seaweed, and vegetables. Decreased iodine intake can cause iodine deficiency and decreased thyroid hormone synthesis. Iodine deficiency can cause cretinism, goiter, myxedema coma, and hypothyroidism.[1][2][3]

Cellular Level

Regulation of thyroid hormone starts at the hypothalamus. The hypothalamus releases thyrotropin-releasing hormone (TRH) into the hypothalamic-hypophyseal portal system to the anterior pituitary gland. TRH stimulates thyrotropin cells in the anterior pituitary to release thyroid-stimulating hormone (TSH). TRH is a peptide hormone created by the cell bodies in the periventricular nucleus (PVN) of the hypothalamus. These cell bodies project their neurosecretory neurons down to the hypophyseal portal circulation, where TRH can concentrate before reaching the anterior pituitary.

TRH is a tropic hormone, meaning that it indirectly affects cells by stimulating other endocrine glands first. It binds to the TRH receptors on the anterior pituitary gland, causing a signal cascade mediated by a G-protein coupled receptor. Activation of Gq protein leads to the activation of phosphoinositide-specific phospholipase C (PLC). PLC hydrolyzes phosphatidylinositol 4,5-P(PIP) into inositol 1,4,5-triphosphate (IP) and 1,2-diacylglycerol (DAG). These second messengers mobilize intracellular calcium stores and activate protein kinase C, leading to downstream gene activation and transcription of TSH. TRH also has a non-tropic effect on the pituitary gland through the hypothalamic-pituitary-prolactin axis. As a non-tropic hormone, TRH directly stimulates lactotropic cells in the anterior pituitary to produce prolactin. Other substances like serotonin, gonadotropin-releasing hormone, and estrogen can also stimulate prolactin release. Prolactin can cause breast tissue growth and lactation.[4]

TSH is released into the blood and binds to the thyroid-releasing hormone receptor (TSH-R) on the basolateral aspect of the thyroid follicular cell. The TSH-R is a Gs-protein coupled receptor, and its activation leads to the activation of adenylyl cyclase and intracellular levels of cAMP. The increased cAMP activates protein kinase A (PKA). PKA phosphorylates different proteins to modify their functions. The five steps of thyroid synthesis are below:

1. **Synthesis of Thyroglobulin:** Thyrocytes in the thyroid follicles produce a protein called thyroglobulin (TG). TG does not contain any iodine, and it is a precursor protein stored in the lumen of follicles. It is produced in the rough endoplasmic reticulum. Golgi apparatus pack it into the vesicles, and then it enters the follicular lumen through exocytosis.
2. **Iodide uptake:** Protein kinase A phosphorylation causes increased activity of basolateral Na⁺-I⁻ symporters, driven by Na⁺-K⁺-ATPase, to bring iodide from the circulation into the thyrocytes. Iodide then diffuses from the basolateral side to the apex of the cell, where it is transported into the colloid through the pendrin transporter.
3. **Iodination of thyroglobulin:** Protein kinase A also phosphorylates and activates the enzyme thyroid peroxidase (TPO). TPO has three functions: oxidation, organification, and coupling reaction.
 1. Oxidation: TPO uses hydrogen peroxide to oxidize iodide (I⁻) to iodine (I₂). NADPH-oxidase, an apical enzyme, generates hydrogen peroxide for TPO.
 2. Organification: TPO links tyrosine residues of thyroglobulin protein with I₂. It generates monoiodotyrosine (MIT) and diiodotyrosine (DIT). MIT has a tyrosine residue with iodine, and DIT has tyrosine residues with 2 molecules of iodine.
 3. Coupling reaction: TPO combines iodinated tyrosine residues to make triiodothyronine (T₃) and tetraiodothyronine (T₄). MIT and DIT join to form T₃, and two DIT molecules form T₄.
4. **Storage:** thyroid hormones are bound to thyroglobulin for storage in the follicular lumen.
5. **Release:** thyroid hormones are released into the fenestrated capillary network by thyrocytes in the following steps:
 1. Thyrocytes uptake iodinated thyroglobulin via endocytosis
 2. Lysosome fuse with the endosome containing iodinated thyroglobulin
 3. Proteolytic enzymes in the endolysosome cleave thyroglobulin into MIT, DIT, T₃, and T₄.
 4. T₃ (20%) and T₄ (80%) are released into the fenestrated capillaries via MCT8 transporter.[5]
 5. Deiodinase enzymes remove iodine molecules from DIT and MIT. Iodine can be salvaged and redistributed to an intracellular iodide pool.[1][6][4]

Organ Systems Involved

Thyroid hormone affects virtually every organ system in the body, including the heart, CNS, autonomic nervous system, bone, GI, and metabolism. In general, when the thyroid hormone

binds to its intranuclear receptor, it activates the genes for increasing metabolic rate and thermogenesis. Increasing metabolic rate involves increased oxygen and energy consumption.

Heart

Thyroid hormones have a permissive effect on catecholamines. It increases the expression of beta-receptors to increase heart rate, stroke volume, cardiac output, and contractility.

Lungs

Thyroid hormones stimulate the respiratory centers and lead to increased oxygenation because of increased perfusion.

Skeletal Muscles

Thyroid hormones cause increased development of type II muscle fibers. These are fast-twitch muscle fibers capable of fast and powerful contractions.

Metabolism

Thyroid hormone increases the basal metabolic rate. It increases the gene expression of Na⁺/K⁺ ATPase in different tissues leading to increased oxygen consumption, respiration rate, and body temperature. Depending on the metabolic status, it can induce lipolysis or lipid synthesis. Thyroid hormones stimulate the metabolism of carbohydrates and anabolism of proteins. Thyroid hormones can also induce catabolism of proteins in high doses. Thyroid hormones do not change the blood glucose level, but they can cause increased glucose reabsorption, gluconeogenesis, glycogen synthesis, and glucose oxidation.

Growth During Childhood

In children, thyroid hormones act synergistically with growth hormone to stimulate bone growth. It induces chondrocytes, osteoblasts, and osteoclasts. Thyroid hormone also helps with brain maturation by axonal growth and the formation of the myelin sheath.[7]

Function

The physiological effects of thyroid hormones are listed below:

- Increases the basal metabolic rate
- Depending on the metabolic status, it can induce lipolysis or lipid synthesis.
- Stimulate the metabolism of carbohydrates
- Anabolism of proteins. Thyroid hormones can also induce catabolism of proteins in high doses.
- Permissive effect on catecholamines
- In children, thyroid hormones act synergistically with growth hormone to stimulate bone growth.
- The impact of thyroid hormone on CNS is important. During the prenatal period, it is needed for the maturation of the brain. In adults, it can affect mood. Hyperthyroidism can lead to hyperexcitability and irritability. Hypothyroidism can cause impaired memory, slowed speech, and sleepiness.

- Thyroid hormone affects fertility, ovulation, and menstruation.

Mechanism

Thyroid hormones are lipophilic and circulate bound to the transport proteins. Only a fraction (approximately 0.2%) of the thyroid hormone (free T4) is unbound and active. Transporter proteins include thyroxine-binding globulin (TBG), transthyretin, and albumin. TBG transports the majority (two-thirds) of the T4, and transthyretin transports thyroxine and retinol. When it reaches its target site, T3 and T4 can dissociate from their binding protein to enter cells either by diffusion or carrier-mediated transport. Receptors for T3 bind are already bound to the DNA in the nucleus before the ligand binding. T3 or T4 then bind to nuclear alpha or beta receptors in the respective tissue and cause activation of transcription factors leading to the activation of certain genes and cell-specific responses. Thyroid hormones are degraded in the liver via sulfation and glucuronidation and excreted in the bile.[8]

Thyroid receptors are transcription factors that can bind to both T3 and T4. However, they have a much higher affinity for T3. As a result, T4 is relatively inactive. Deiodinases convert T4 to active T3 or inactive reverse T3 (rT3). There are three types of deiodinases: type I, II, and III. Type I (DIO1) and II (DIO2) are located in the liver, kidneys, muscles, and thyroid glands. Type III (DIO3) deiodinases are located in the CNS and placenta. DIO1 and DIO2 convert T4 to the active form T3, and DIO3 converts T4 into the inactive form rT3.[8]

Related Testing

Hypothalamus releases thyrotropin-releasing hormone (TRH) that stimulates the secretion of TSH in the pituitary gland. Increased free T4 and T3 inhibit the release of TRH and TSH through a negative feedback loop. As a result, T3 and T4 secretion and iodine uptake are reduced. Other hormones, such as somatostatin, glucocorticoids, and dopamine, also inhibit TSH production. Cold, stress, and exercise increase TRH release.

The initial tests of choice to screen for any thyroid abnormality are a TSH and a free thyroxine (free T4) test. These determine whether the abnormality arises centrally from the thyroid gland (primary), peripherally from the pituitary (secondary), or hypothalamus (tertiary). In primary hypothyroidism is suspected, the thyroid gland is not releasing enough thyroid hormones. Therefore, TSH levels will be appropriately elevated, while free T4 levels will be lower. In primary hyperthyroidism, free T4 levels are abnormally increased, and TSH levels will be appropriately decreased. Other lab tests such as TSH receptor antibodies or antibodies to thyroid peroxidase can help aid in diagnosing Graves disease or Hashimoto thyroiditis, respectively.[9]

In pregnant women, thyroid-binding globulin production is increased because of estrogen and beta-human chorionic gonadotropin (beta-HCG). More free T4 will be bound to TGB, leading to increased production of T4. TSH levels and free T4 levels will normalize, and total T4 will increase. Therefore, laboratory values will show normal TSH, normal free T4, and elevated total T4.[10]

Pathophysiology

Hyperthyroidism: Disorders of the thyroid gland can result in excess T3 and T4 production along with the compensatory decrease of TSH. In addition, thyrotroph adenoma can produce unregulated TSH and can lead to increased T3 and T4 production. There is an ectopic production

of thyroid hormone in some conditions, leading to increased thyroid hormones and compensatory TSH decrease.

Hypothyroidism: In primary hypothyroidism, decreased production of thyroid hormones by the thyroid gland causes a compensatory increase of TSH. Secondary hypothyroidism is caused by pituitary disorders causing decreased TSH release and decreased T3/T4 levels. Tertiary hypothyroidism is caused by hypothalamic disorders, resulting in decreased TRH levels, decreased TSH, and T3/T4 levels.

Two of the most common causes of hyperthyroidism and hyperthyroidism are below:

Graves Disease

Graves disease is the most common cause of hyperthyroidism. It is an autoimmune disease caused by the production of TSH receptor antibodies that stimulate thyroid gland growth and thyroid hormone release. Patients will have abnormally increased T4 and T3 levels and a decrease in TSH. A positive TSH-receptor IgG immunoglobulin test confirms the diagnosis. Immunoglobulin G (IgG) against TSH-receptor leads to increased thyroid function and growth. Patients will often present with symptoms of hyperthyroidism and diffuse goiter. TSH-receptor antibodies can also activate orbital fibroblasts leading to fibroblast proliferation and differentiation to adipocytes. As a result, there is increased production of hyaluronic acid and glycosaminoglycan (GAG), leading to an increased volume of intraorbital fat and muscle tissue. It causes exophthalmos, lid retraction, and diplopia due to ocular motility problems. Pretibial myxedema is another finding in Graves' disease. It is due to the stimulation of dermal fibroblasts that leads to depositions of GAGs in the connective tissue. 70% of patients with Graves disease have elevated anti-TPO antibodies.

Hashimoto Thyroiditis

The most common cause of hypothyroidism in iodine-sufficient areas is Hashimoto Thyroiditis. It is caused by autoimmune-mediated destruction of the thyroid gland. CD8+ T-cells cause thyroid follicular cell death. The release of IFN-gamma by TH1 cells causes the recruitment and activation of macrophages. During the early stage of the disease, the patient may develop a non-tender, symmetrical, and painless goiter. As inflammation continues, thyroid follicles are damaged and can rupture. When thyroid follicles rupture, the patient may be asymptomatic or can experience Hashitoxicosis (thyroid hormone from ruptured follicles, causing symptoms of hyperthyroidism). As the disease progresses, the thyroid gland may become normal-size or small, depending on the extent of fibrosis. As a result, the patient can develop the symptoms of hypothyroidism. In addition to cell-mediated destruction, anti-thyroid autoantibodies (anti-thyroglobulin and anti-TPO) are also produced, leading to antibody-dependent cell-mediated cytotoxicity. Hashimoto thyroiditis is diagnosed via ultrasound, antibody detection, and thyroid function testing. Radioactive iodine uptake test and fine-needle aspiration can be performed to exclude malignancy.

Clinical Significance

Symptoms of Hypothyroidism

Generalized decreased basal metabolic rate can present as apathy, slowed cognition, skin dryness, alopecia, increased low-density lipoproteins, and increased triglycerides.

Hypothyroidism must be ruled out in psychiatry patients presenting with apathy and slowed cognition. Hypothyroidism can decrease sympathetic activity leading to decreased sweating,

bradycardia, and constipation. Patients can present with myopathy and decreased cardiac output because of decreased transcription of sarcolemmal genes.

Hyperprolactinemia can be caused by hypothyroidism. Thyrotropin-releasing hormone (TRH) from the hypothalamus stimulates prolactin and TSH release. Prolactin release can suppress testosterone, LH, FSH, and GnRH release. Prolactin can also cause breast tissue growth.

Patients with hypothyroidism may present with myxedema caused by decreased clearance of complex glycosaminoglycans and hyaluronic acids from the reticular layer of the dermis. Initially, the nonpitting edema is pretibial. As the state of hypothyroidism continues, patients can develop generalized edema.

Symptoms related to decreased metabolic rate:

- Bradycardia
- Fatigue
- Cold intolerance
- Weight gain
- Poor appetite
- Hair loss
- Cold and dry skin
- Constipation
- Myopathy, stiffness, cramps, entrapment syndromes
- Delayed deep tendon reflex relaxation

Symptoms from generalized myxedema:

- Myxedematous heart disease
- Puffy appearance with doughy skin texture
- Hoarse voice with difficulty articulating words
- Pretibial and periorbital edema

Symptoms of hyperprolactinemia:

- Amenorrhea or menorrhagia
- Galactorrhea
- Erectile dysfunction, infertility in men
- Decreased libido

Other symptoms:

- Depression
- Impaired concentration and memory

- Goiter
- Hypertension

Congenital hypothyroidism:

- Umbilical hernia
- Hypotonia
- Prolonged neonatal jaundice
- Poor feeding, absence of thirst (adipsia)
- Decreased activity
- Pot-belly, puffy-face, protuberant tongue
- Poor brain development

Symptoms of Hyperthyroidism

Generalized hypermetabolism from hyperthyroidism causes increased $\text{Na}^+/\text{K}^+-\text{ATPase}$ to promote thermogenesis. There is increased catecholamine secretion, and beta-adrenergic receptors are also upregulated in various tissues. As a result of the hyperadrenergic state, peripheral vascular resistance is decreased. In the heart, hyperthyroidism causes a decreased amount of phospholamban, a protein that normally decreases the affinity of calcium-ATPase for calcium in the sarcoplasmic reticulum. As a result of decreased phospholamban, there is increased Ca^{2+} movement between the sarcoplasmic reticulum and cytosol, leading to increased contractility. Increased beta receptors in the heart also lead to increased cardiac output.

General

- Heat intolerance
- Weight loss
- Increased appetite
- Increased sweating from cutaneous blood flow increase
- Weakness
- Fatigue
- Onycholysis (separation of nails from nail beds)
- Pretibial myxedema

Eyes

- Lid lag (when looking down, sclera visible above cornea)
- Lid retraction (when looking straight, sclera visible above the cornea)
- Graves ophthalmopathy

Goiter

- Diffuse, smooth, non-tender goiter
- The audible bruit can be heard at the superior poles

Cardiovascular

- Tachycardia (can be masked by patients taking beta-blockers)
- Palpitations
- An irregular pulse from atrial fibrillation
- Hypertension
- Widened pulse pressure because systolic pressure increases and diastolic pressure decreases
- Heart failure (elderly patients)
- Chest pain
- Abnormal heart rhythms

Musculoskeletal

- Fine tremors of the outstretched fingers. Face, tongue, and head can also be involved. Tremors respond well to treatment with beta-blockers.
- Myopathy affecting proximal muscles. Serum creatine kinase levels can be normal.
- Osteoporosis is caused by the direct effects of T3. Older patients can present with fractures.

Neuropsychiatric system

- Restlessness
- Anxiety
- Depression
- Emotional instability
- Insomnia
- Tremulousness
- Hyperreflexia

Conditions associated with hypothyroidism

- Iodine deficiency [11]
- Cretinism [12]
- Wolff-Chaikoff effect [13]
- Subacute thyroiditis [14]
- Postpartum thyroiditis [15]

- Riedel thyroiditis [16]
- Hashimoto thyroiditis [17]
- Drug-induced [18]

Conditions associated with hyperthyroidism

- Graves disease [19]
- Iodine excess [20]
- Struma ovarii [21]
- Thyrotropic pituitary adenoma [22]
- Jod-Basedow phenomenon [23]
- Drug-induced: amiodarone, lithium [24]
- Thyrotoxicosis and thyroid storm [25]
- Toxic multinodular goiter [26]
- Thyroid adenoma [27]

Antithyroid drugs that work in the thyroid gland [28]

- Perchlorate – inhibits Na⁺/I⁻ symporter – blocks iodide uptake
- Thionamides – inhibits TPO – block thyroid hormone synthesis
- Iodide > 5mg – inhibits Na⁺/I⁻ symporter and TPO – blocks iodide uptake and thyroid hormone synthesis
- Lithium – inhibits thyroid hormone release (off-label use for thyroid storm)

Antithyroid drugs that work in peripheral tissue – all these drugs inhibit the deiodinase enzymes. Deiodinase enzymes normally convert T4 into the active form T3. These drugs inhibit the conversion of T4 to T3 and reduce its activity.

- Propylthiouracil (thionamide)
- Dexamethasone
- Amiodarone
- Propranolol

Review Questions

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Disclosure: Muhammad Shahid declares no relevant financial relationships with ineligible companies.

Disclosure: Muhammad Ashraf declares no relevant financial relationships with ineligible companies.

Disclosure: Sandeep Sharma declares no relevant financial relationships with ineligible companies.

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Bookshelf ID: NBK500006 PMID: 29763182